

1.1 Physical Quantities & Units

Question Paper

Course	CIE A Level Physics
Section	1. Physical Quantities & Units
Topic	1.1 Physical Quantities & Units
Difficulty	Easy

Time allowed: 50

Score: /40

Percentage: /100

Question 1a

Distinguish between scalar and vector quantities.

[2 marks]

Question 1b

(i)

Energy is a scalar quantity. State **three** other scalar quantities.

[3]

(ii)

Force is a vector quantity. State **three** other vector quantities.

[3]

[6 marks]

Question 1c

Fig. 1.1 shows a block at rest on a slope.

Three forces, weight W , normal contact force N and the frictional force F act on the block. The forces are in equilibrium.

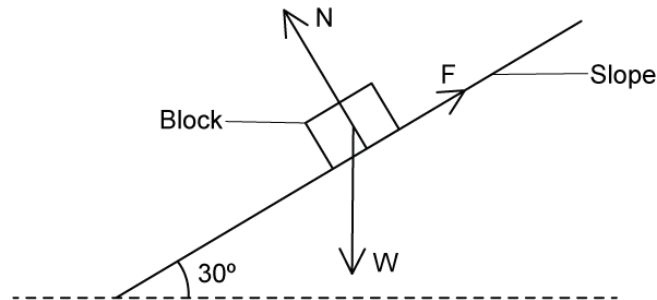


Fig. 1.1 (not to scale)

(i)

Explain what is meant by 'the forces are in equilibrium'.

[2]

(ii)

Complete the vector triangle in Fig. 1.2. Label the missing vector and the angle between W and N .

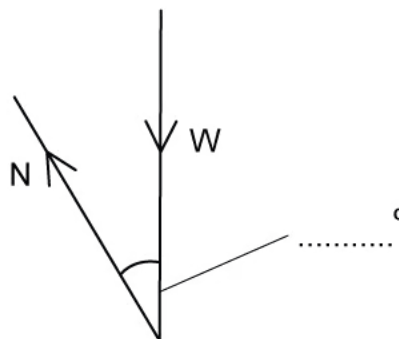


Fig. 1.2

[2]

[4 marks]

Question 1d

The frictional force has a magnitude of 60 N.

By resolving the vector, calculate the component of the force that acts

(i)
horizontally

[2]

(ii)
vertically.

[2]

[4 marks]

Question 2a

Table 1.1 shows SI base quantities and their corresponding base unit.

Table 1.1

SI base quantity	SI base unit
Mass	kilogram, kg
Time	
Length	

(i)

Complete the missing SI base units for time and length in Table 1.1.

[2]

(ii)

Complete Table 1.1 by adding two other base quantities and their units.

[2]

[4 marks]

Question 2b

Newton's second law of motion describes the relationship between force F mass m and acceleration a :

$$F = ma$$

The derived unit for force is the newton (N).

Use Newton's second law to write 1 N in SI units.

[2 marks]

Question 2c

The kinetic energy equation describes the relationship between energy E , mass, m , and velocity, v .

$$E_k = \frac{1}{2}mv^2$$

The derived unit for energy is the joule (J).

Use the equation linking kinetic energy and speed to write 1 J in SI base units.

[2 marks]**Question 2d**

The derived unit for power P is the watt W.

Given that power is the energy transferred E per unit time t , deduce the SI base units for power

[2 marks]**Question 3a**

On Fig. 1.1, draw a line from each quantity to its correct unit.

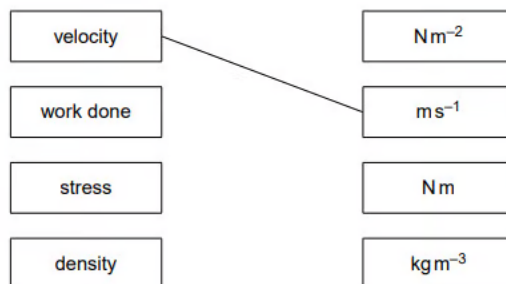


Fig. 1.1

[3 marks]

Question 3b

Complete the missing prefixes and powers of ten in Table 1.1

Table 1.1

Prefix	Power of ten
tera	
	10^9
	10^3
centi	10^{-2}
milli	
micro	
	10^{-9}
pico	10^{-12}

[3 marks]

Question 3c

Convert the following measurements into standard form and to the specified number of significant figures.

(i) 71 GJ to 3 s.f.

[1]

(ii)
24.5 kW to 4 s.f.

[1]

(iii)
0.003823 mm to 1 s.f.

[1]

(iv)
644.9 pN to 2 s.f.

[1]

[4 marks]**Question 3d**

Estimate the orders of magnitude with an appropriate SI unit and correct prefix for the quantities in Table 1.2

Table 1.2

Quantity	Order of magnitude
Mass of a car	
Thickness of a sheet of paper	
Density of air (at room temperature and pressure)	
Time between two heart beats	

[4 marks]

Model Answers: Easy

1a

(a) The difference between scalars and vectors is:

- Scalar quantities have magnitude / size only; [1 mark]
- Vector quantities have both magnitude / size and direction; [1 mark]

[Total: 2 marks]

1b

(i) Examples of scalar quantities:

Any **three** from:

- Distance; [1 mark]
- Speed; [1 mark]
- Time; [1 mark]
- Mass; [1 mark]
- Power; [1 mark]
- Temperature; [1 mark]
- Electrical quantities e.g. resistance / current / potential difference; [1 mark]

(ii) Examples of vector quantities:

Any **three** from:

- Displacement; [1 mark]
- Velocity; [1 mark]
- Acceleration; [1 mark]
- Momentum; [1 mark]
- Moments / torque; [1 mark]

[Total: 6 marks]

In part (i), answers such as 'kinetic energy' or 'gravitational potential energy' would not score a mark as these are **types** of energy

In part (ii), answers such as 'weight' or 'tension' would not score a mark as these are **types** of forces

You might think that current and potential difference **could** be vectors because they can have both magnitude and direction. However, they **cannot** be added together or resolved **as vectors** in the same way that force and velocity can, so they are classed as scalar quantities

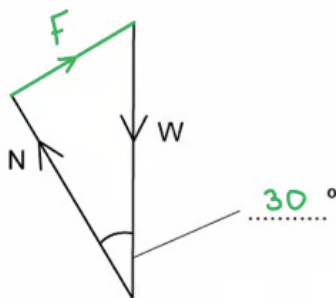
1c

(i) The 'forces are in equilibrium' means:

Any **two** from:

- There are no resultant / net forces (acting on the block); [1 mark]
- All the forces are balanced (so the block doesn't move); [1 mark]
- The forces form a closed vector triangle; [1 mark]

(ii) The completed vector triangle is as follows:



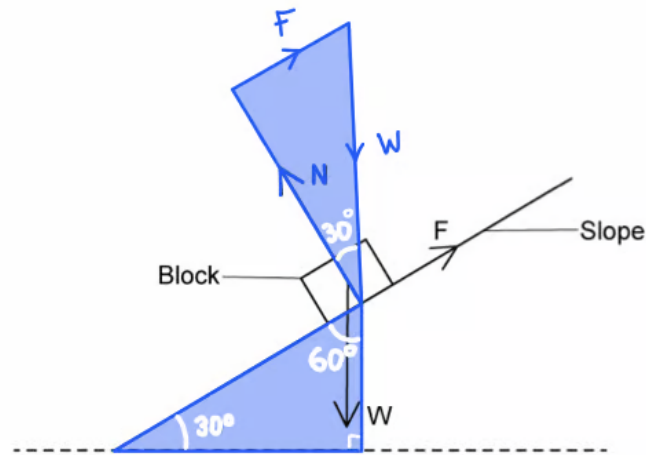
- Frictional force F correctly drawn and labelled; [1 mark]
- Angle labelled 30° ; [1 mark]

[Total: 4 marks]

When drawing forces in **equilibrium**, the arrows should always form a complete triangle with arrows matching head-to-tail

The geometry of triangles and angle addition rules shown below outline why the angle

between N and W is the same as the angle of inclination of the slope



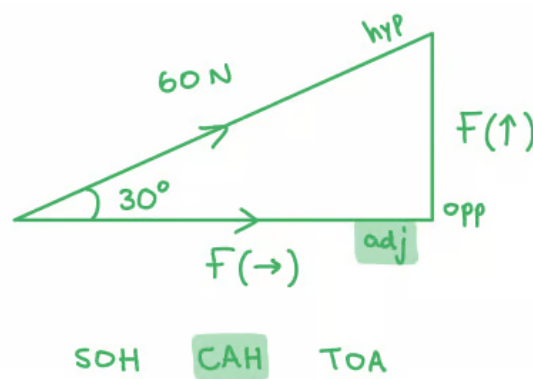
1d

(d)

List the known quantities:

- Frictional force, $F = 60 \text{ N}$
- Angle between F and the horizontal $= 30^\circ$

(i) Resolving the vector horizontally:

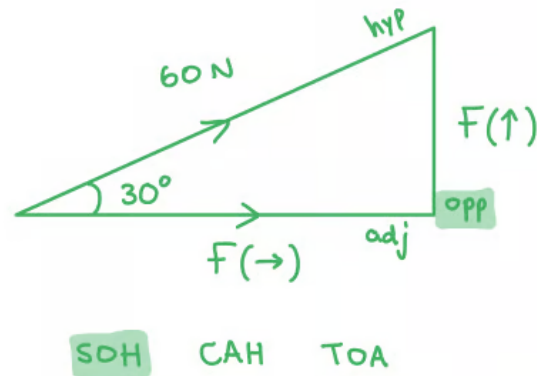


$$\bullet \cos 30^\circ = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{F(\rightarrow)}{60}$$

$$\bullet F(\rightarrow) = 60 \times \cos 30^\circ \text{ [1 mark]}$$

$$\bullet F(\rightarrow) = 51.96 = 52 \text{ N [1 mark]}$$

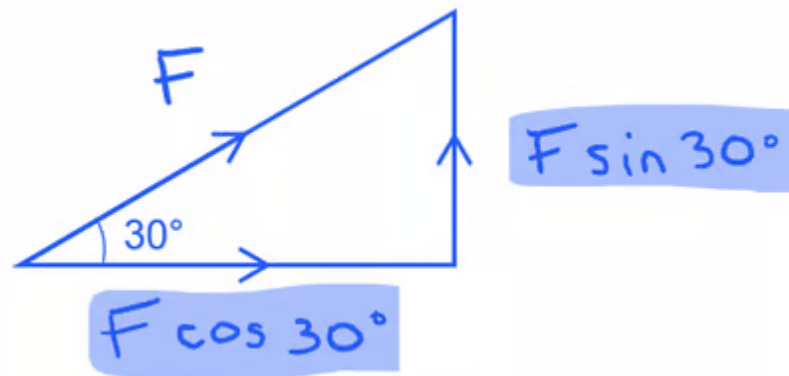
(ii) Resolving the vector vertically:



- $\sin 30^\circ = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{F(\uparrow)}{60}$
- $F(\uparrow) = 60 \times \sin 30^\circ$ [1 mark]
- $F(\uparrow) = 30 \text{ N}$ [1 mark]

[Total: 4 marks]

Drawing the triangles is essential to answering this question correctly.



2a

(a)

(i) The SI base units are:

SI base quantity	SI base unit



2a

(a)

(i) The SI base units are:

SI base quantity	SI base unit
Mass	kilogram, kg
Time	second, s
Length	metre, m

- Length = metre / m; [1 mark]
- Time = second / s; [1 mark]

(ii) Two other SI base quantities & their units are:

Any **two** from:

- Temperature **AND** Kelvin / K; [1 mark]
- Current **AND** amp / ampere / A; [1 mark]
- Amount of substance **AND** mole / mol; [1 mark]

[Total: 4 marks]

2b

(b) Write 1 N in SI base units:

- $F = ma \Rightarrow [F] = [m][a]$
 - $[m] = \text{kg}$
 - $[a] = \text{m s}^{-2}$
- $1 \text{ N} = (\text{kg}) \times (\text{m s}^{-2})$ [1 mark]
- $1 \text{ N} = \text{kg m s}^{-2}$ [1 mark]

[Total: 2 marks]

It is really important that you learn the SI base units in case you are asked a question like this in the exam

Once you know the SI base units, it is fairly straightforward to deduce the base units for any derived unit because the method is exactly the same as the method given here.

2c

(c) Write 1 J in SI base units:

- $E_k = \frac{1}{2}mv^2 \Rightarrow [E_k] = [m][v]^2$
 - $[m] = \text{kg}$
 - $[v] = \text{m s}^{-1}$
- $1 \text{ J} = (\text{kg}) \times (\text{m s}^{-1})^2$ [1 mark]
- $1 \text{ J} = \text{kg m}^2 \text{ s}^{-2}$ [1 mark]

[Total: 2 marks]

The same method is used to determine the SI base units for any derived unit

Practice this skill as part of your revision. It will help you to remember the SI base units (if you don't already know them) rather than having to learn them by rote.

2d

(d) Deduce the SI base unit for W:

- Power: $P = \frac{E}{t} \Rightarrow [P] = \frac{[E]}{[t]}$
 - $[E] = \text{kg m}^2 \text{ s}^{-2}$ (from (c))
 - $[t] = \text{s}$
- $W = \frac{\text{kg m}^2 \text{ s}^{-2}}{\text{s}}$ [1 mark]
- $W = \text{kg m}^2 \text{ s}^{-3}$ [1 mark]

[Total: 2 marks]

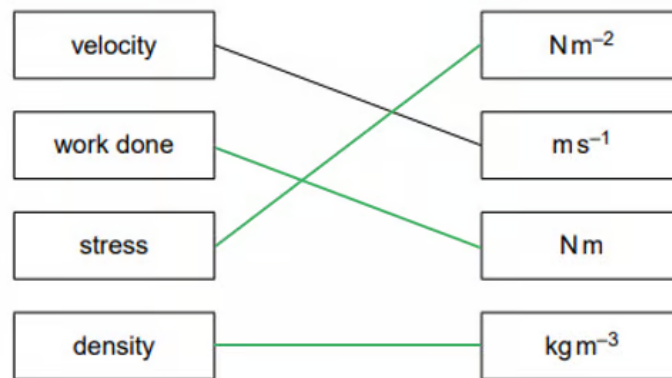
It would be a good idea to practice deducing the SI base units for a variety of derived

units so that you get comfortable with the method and familiar with the SI units

Try it for each topic as part of your revision.

3a

(a) The correct quantities and units are as follows:



- Work done = Nm [1 mark]
- Stress = Nm^{-2} [1 mark]
- Density = kgm^{-3} [1 mark]

[Total: 3 marks]

Make sure you can recall equations in order to derive SI units as quickly as possible

In this case, you should be able to quickly recall:

$$\text{Work done} = \text{Force} \times \text{distance}$$

$$\text{[Work done]} = (\text{N}) \times (\text{m}) = \text{Nm}$$

$$\text{Stress} = \text{Force} / \text{area}$$

$$\text{[Stress]} = (\text{N}) / (\text{m}^2) = \text{Nm}^{-2}$$

$$\text{Density} = \text{Mass} / \text{volume}$$

$$\text{[Density]} = (\text{kg}) / (\text{m}^3) = \text{kgm}^{-3}$$

3b

(b) A correctly table of prefixes and powers of ten is as follows:

Prefix	Power of ten
tera	10^{12}
giga	10^9
kilo	10^3
centi	10^{-2}
milli	10^{-3}
micro	10^{-6}
nano	10^{-9}
pico	10^{-12}

- 3 marks for all six correct
- 2 marks for four or five correct
- 1 mark for two or three correct
- 0 marks for less than two correct

[Total: 3 marks]

3c

(c)

(i) 0.71 GJ in standard form and 3 s.f. is:

- $1 \text{ GJ} = 1 \times 10^9 \text{ J}$
- $0.71 \text{ GJ} = 0.71 \times (1 \times 10^9 \text{ J})$
- $0.71 \text{ GJ} = 7.10 \times 10^8 \text{ J} (3 \text{ s.f.})$ [1 mark]

(ii) 24.5 kW in standard form and 4 s.f. is:

- $1 \text{ kW} = 1 \times 10^3 \text{ W}$
- $24.5 \text{ kW} = 24.5 \times (1 \times 10^3 \text{ W})$
- $24.5 \text{ kW} = 2.450 \times 10^4 \text{ W} (4 \text{ s.f.})$ [1 mark]

- 3 marks for all six correct
- 2 marks for four or five correct
- 1 mark for two or three correct
- 0 marks for less than two correct

[Total: 3 marks]

3c

(c)

(i) 0.71 GJ in standard form and 3 s.f. is:

- $1 \text{ GJ} = 1 \times 10^9 \text{ J}$
- $0.71 \text{ GJ} = 0.71 \times (1 \times 10^9 \text{ J})$
- $0.71 \text{ GJ} = 7.10 \times 10^8 \text{ J}$ (3 s.f.) [1 mark]

(ii) 24.5 kW in standard form and 4 s.f. is:

- $1 \text{ kW} = 1 \times 10^3 \text{ W}$
- $24.5 \text{ kW} = 24.5 \times (1 \times 10^3 \text{ W})$
- $24.5 \text{ kW} = 2.450 \times 10^4 \text{ W}$ (4 s.f.) [1 mark]

(iii) 0.003823 mm in standard form and 1 s.f. is:

- $1 \text{ mm} = 1 \times 10^{-3} \text{ m}$
- $0.003823 \text{ mm} = 0.003823 \times (1 \times 10^{-3} \text{ m})$
- $0.003823 \text{ mm} = 4 \times 10^{-6} \text{ m}$ (1 s.f.) [1 mark]

(iv) 644.9 pN in standard form and 2 s.f. is:

- $1 \text{ pN} = 1 \times 10^{-12} \text{ N}$
- $644.9 \text{ pN} = 644.9 \times (1 \times 10^{-12} \text{ N})$
- $644.9 \text{ pN} = 6.4 \times 10^{-10} \text{ N}$ (2 s.f.) [1 mark]

[Total: 4 marks]

You should be familiar with using scientific notation from IGCSE, but it can be tricky when you are first learning. Practice makes perfect.

Make sure you are confident in using and converting numbers in standard form well before your exams. And make sure you know how to quote numbers to appropriate

numbers of significant figures, as this is something that can cost you marks in an exam if you don't do it correctly!

3d

(d) Estimates of the orders of magnitude of the quantities are:

Quantity	Order of magnitude	
Mass of a car	$\sim 1 \times 10^3 \text{ kg}$	[1 mark]
Thickness of a sheet of paper	$\sim 1 \times 10^{-1} \text{ mm}$	[1 mark]
Density of air (at room temperature and pressure)	$\sim 1 \times 10^0 \text{ kg m}^{-3}$	[1 mark]
Time between two heart beats	$\sim 1 \text{ s}$	[1 mark]

Acceptable ranges:

- Mass of a car = 500 → 2000 kg
- Thickness of a sheet of paper = 0.05 mm → 0.5 mm **OR** 50 → 500 μm
- Density of air = 0.1 kg m^{-3} → 5 kg m^{-3}
- Time between two heart beats = 600 → 2000 ms **OR** 0.6 → 2.0 s

[Total: 4 marks]

You should know that the mass of a car is of the order of 1 ton which is 1000 kg

You can estimate the thickness of a sheet of paper by considering a stack of 500 sheets (think about the stacks of paper that you can buy for printers), these are about 1 inch thick which is 2.5 cm or 25 mm, so 1 sheet of paper is about $25 / 500 = 0.05 \text{ mm}$ thick

You should also know that the density of air is about 1 kg m^{-3}

You can consider a heartbeat and the number of beats per minute to calculate the time

between two beats, it's roughly 1 s depending on your resting heart rate

